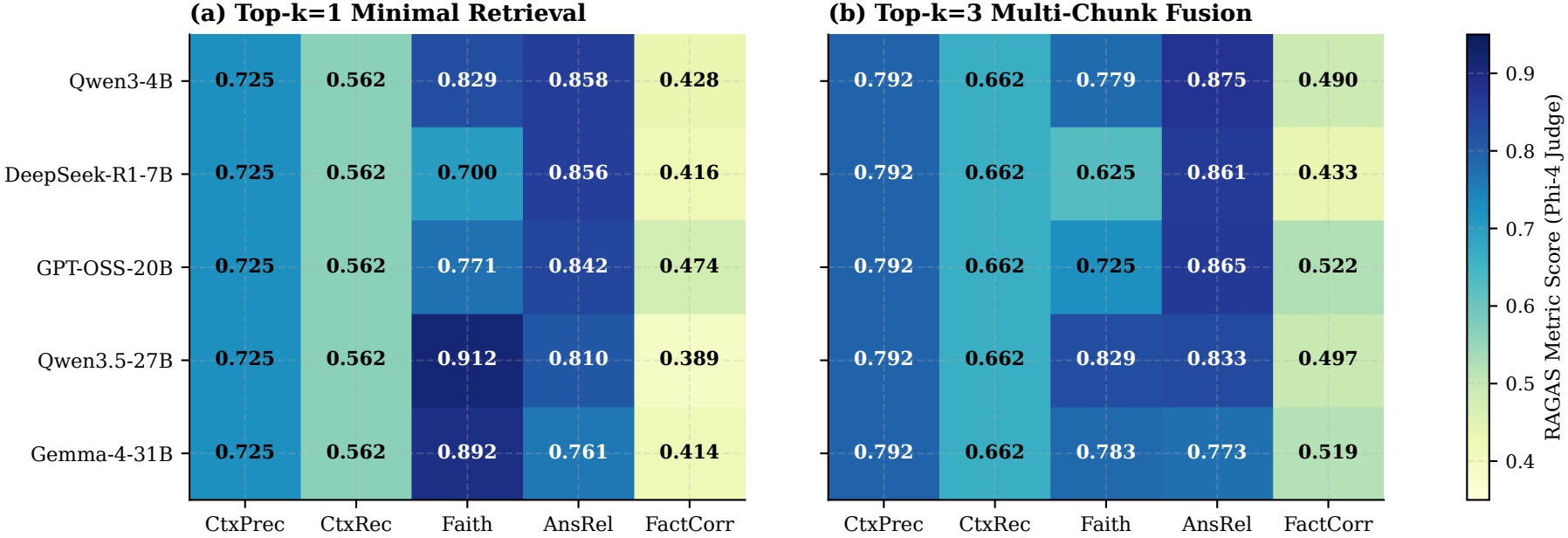
(a) Top-k=1 Minimal Retrieval
(b) Top-k=3 Multi-Chunk Fusion
Qwen3-4B
DeepSeek-R1-7B
GPT-OSS-20B
Qwen3.5-27B
Gemma-4-31B
0.725 0.562 0.829 0.858 0.428
0.725 0.562 0.700 0.856 0.416
0.725 0.562 0.771 0.842 0.474
0.725 0.562 0.912 0.810 0.389
0.725 0.562 0.892 0.761 0.414
0.792 0.662 0.779 0.875 0.490
0.792 0.662 0.625 0.861 0.433
0.792 0.662 0.725 0.865 0.522
0.792 0.662 0.829 0.833 0.497
0.792 0.662 0.783 0.773 0.519
CtxPrec
CtxRec
Faith
AnsRel
FactCorr
RAGAS Metric Score (Phi-4 Judge)
0.9
0.8
0.7
0.6
0.5
0.4